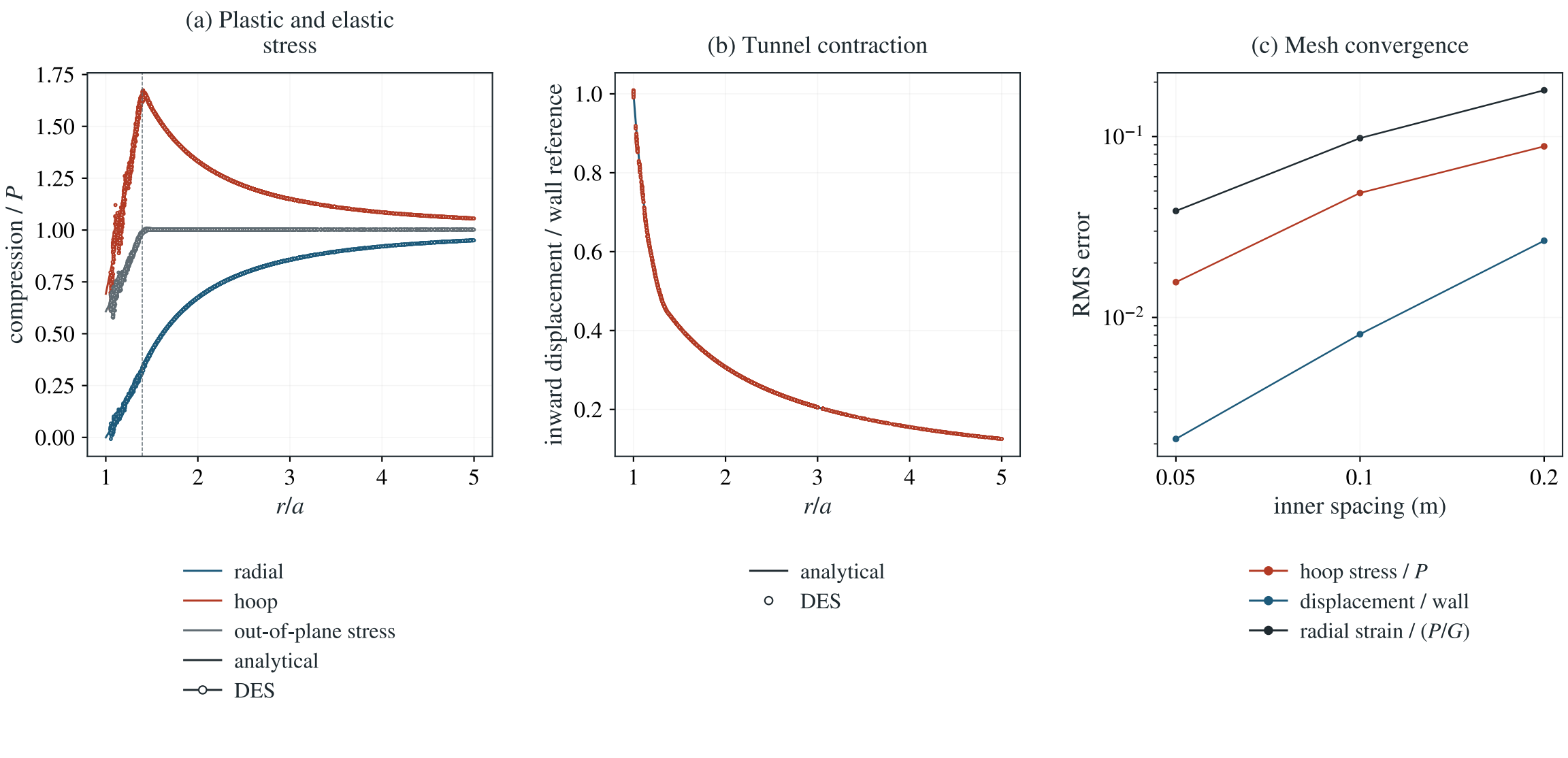

(a) Plastic and elastic stress
compression / P
r/a
radial
hoop
out-of-plane stress
analytical
DES
(b) Tunnel contraction
inward displacement / wall reference
r/a
analytical
DES
(c) Mesh convergence
RMS error
inner spacing (m)
hoop stress / P
displacement / wall
radial strain / (P/G)